

A promotional image for the movie 'The Long Walk Home'. It features the five main cast members standing behind a tank in a post-apocalyptic setting. The background is a hazy, yellowish sky with smoke or dust. The tank is a military vehicle with a long barrel and tracks. The cast members are dressed in military-style clothing. The text 'From the first (data) soldier to the last (report) stand' is overlaid in a large, bold, black font.

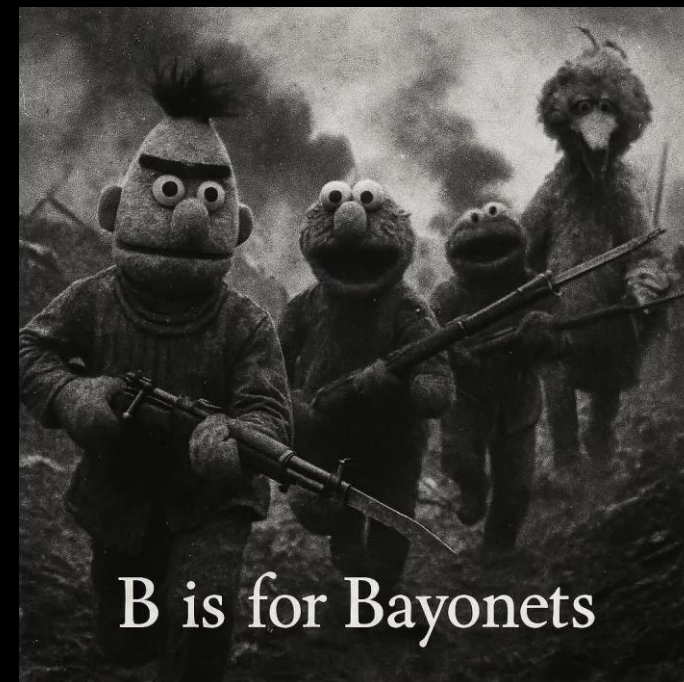
**From the first (data)
soldier to the last
(report) stand**

A sabaton report all the way

Jo David



Sabatón





We are Sparkle

Your holistic data partner

- ⚡ We integrate all relevant areas of expertise into a solution that brings value
- ⚡ We combine in-depth knowledge about data (technology), with the skills to address business challenges
- ⚡ We focus on long term partnerships with our clients and experts

People

50+ experts - Average **+15 years** of experience

Values and culture

Enthusiasm, Humility and Teamwork

Offices

BENELUX - Leuven (HQ/B), Haarlem (NL),
Louvain-la-neuve (B), Hasselt (B), Antwerp (B)

Estonia – Tallinn

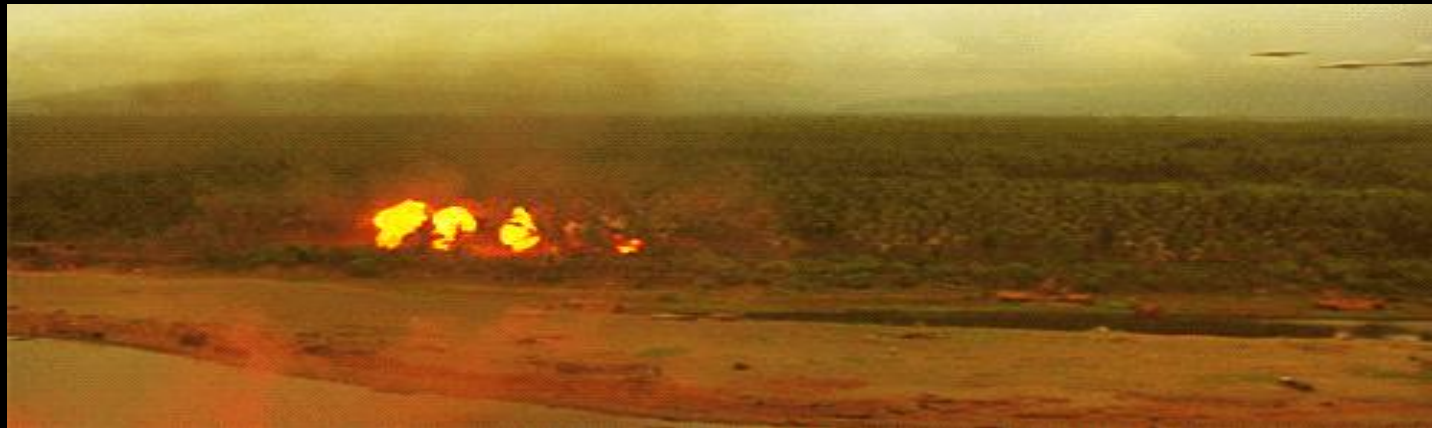
Part of the Cronos Group,

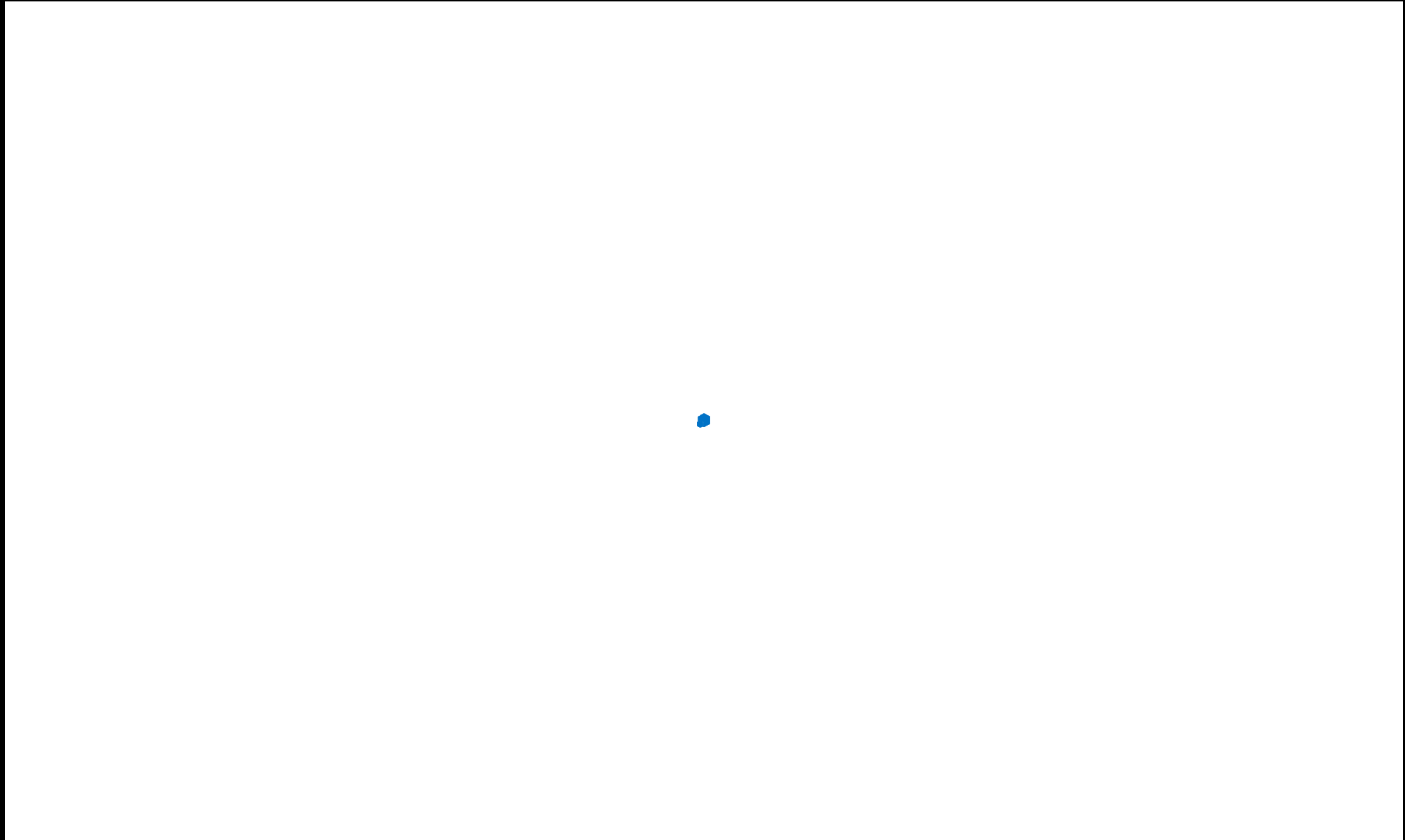
9000+ experts in Europe





website is a great starting point. Like true 'Swedish Pagans',
when we first grab it, so we dive 'Into the fire' and start cleaning
'Back in control', it's time to search for more data to enrich our dataset.





What is Power BI

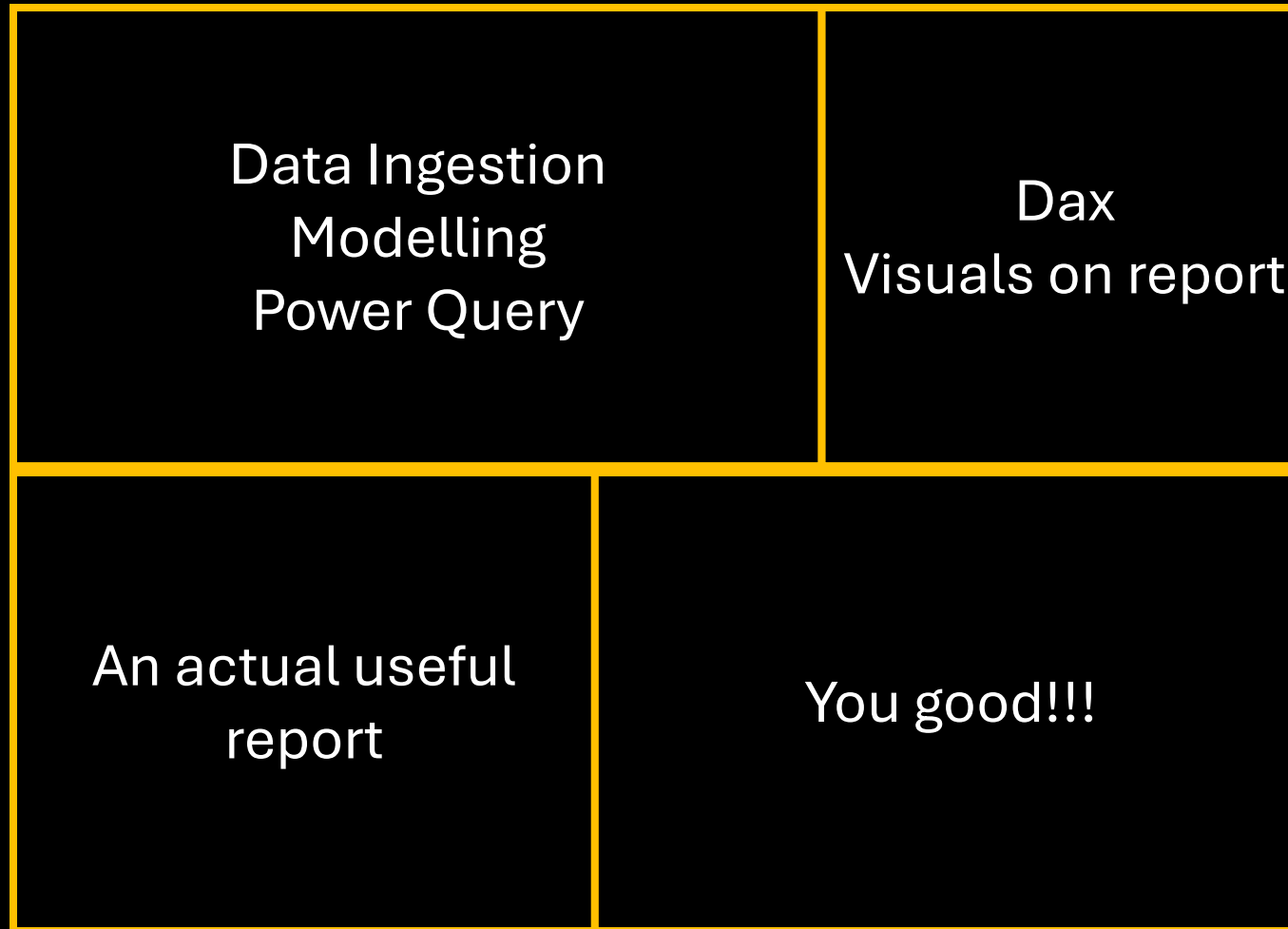
Acc

- analytics platform that helps you
Whether you're a business user,
offers integrated tools and
share data across your

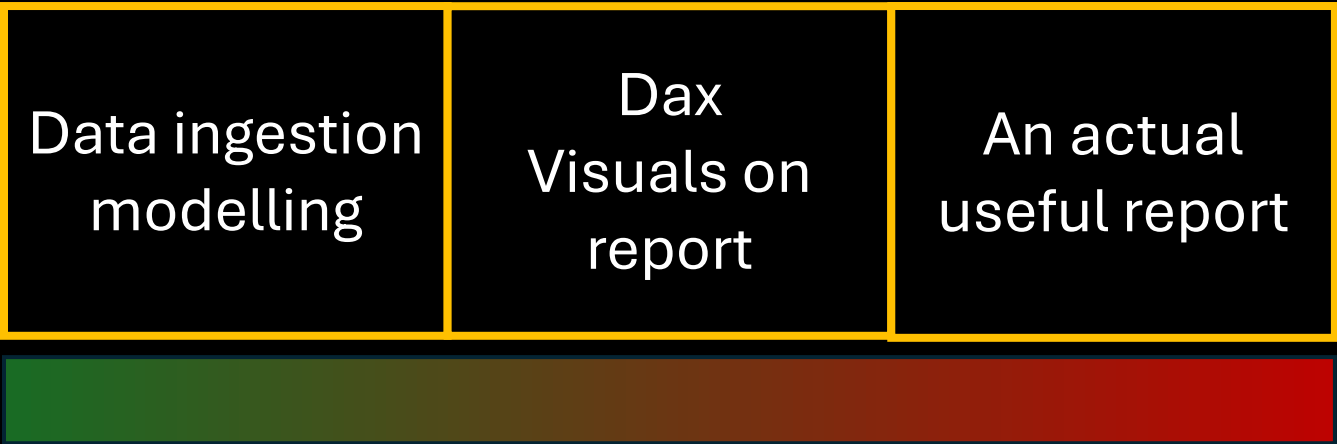
kingdom come

TO MUCH TEXT

What is Power BI?

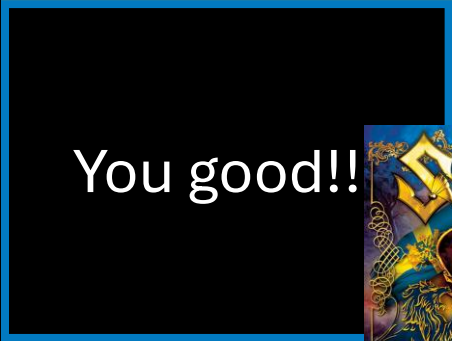


What is Power BI

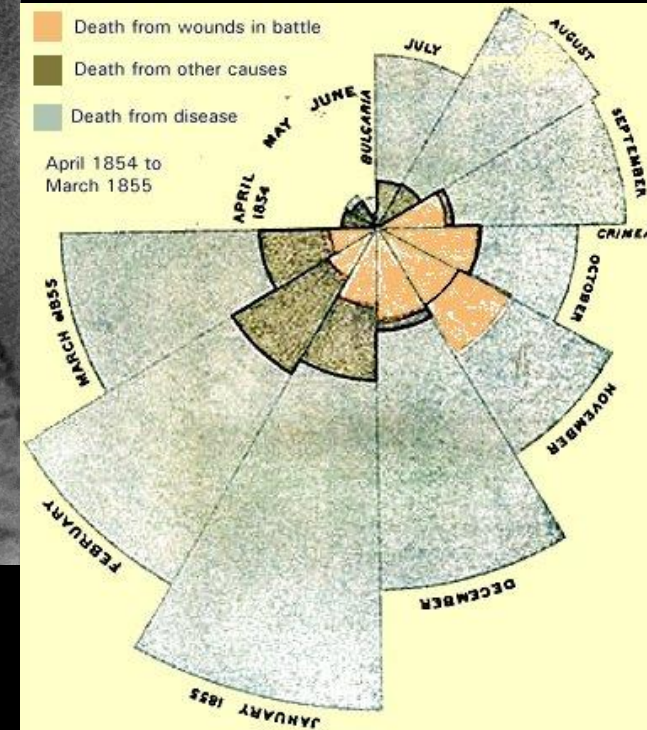
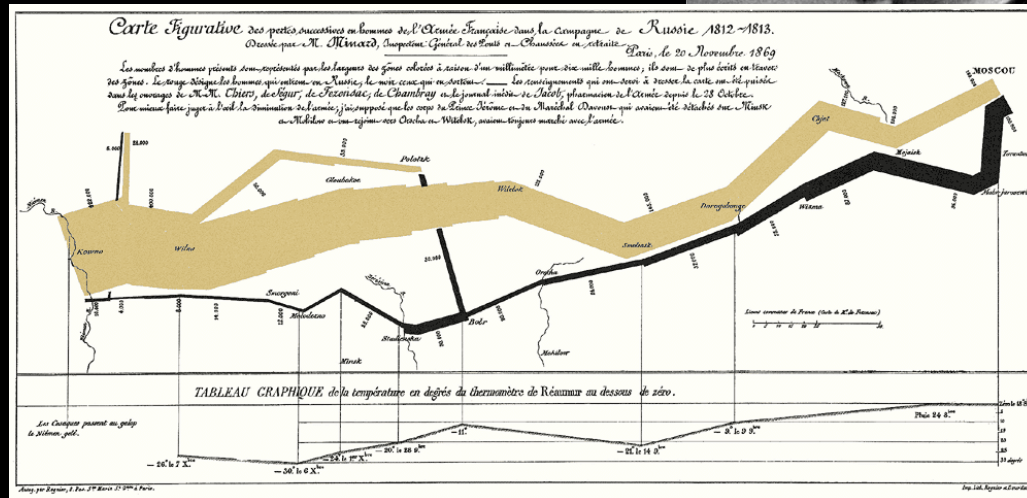


Legend

Tech people like it
Business people like it
I was chosen by heaven

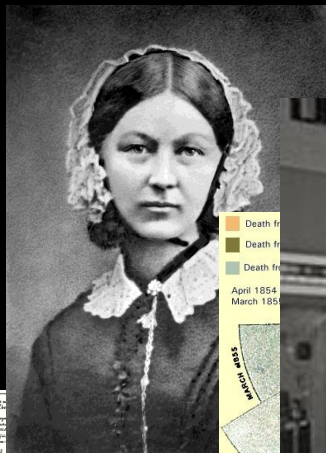


Individuals doing Data



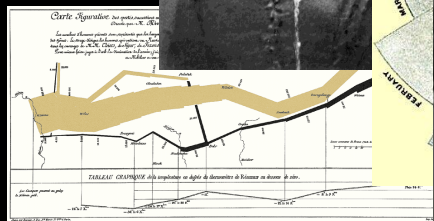


Team work makes the dream work



Death fr
Death fr
Death fr
April 1854
March 185

1854
March 185



Team work makes the dream work



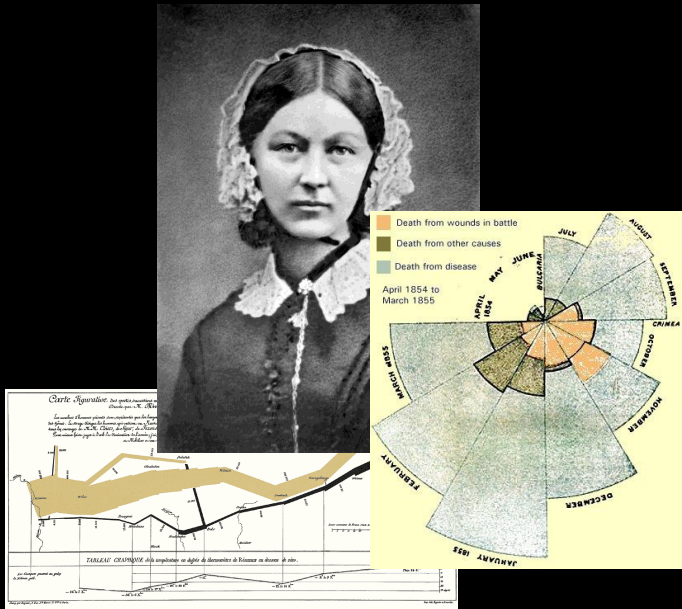
Specialised teams in a larger part



Specialised teams in a larger part



Specialised teams in a larger part



We are BI-troopers



Get data – [Sabaton.net/discography](https://www.sabaton.net/discography)

Step 1 of every report: Data

Where to get it?

How to ingest it?

Quality?

...

Get data – Sabaton.net/discography

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DISCOGRAPHY

[Studio Releases](#)[Live Releases](#)[Playlists](#)

Get data – Best practise warning

- Data is never clean
- Data is hardly ever
- Getting data from
- Ideally you own
- Star schema all
- Use int keys over
- Use implicit me
- Real estate on a
- Hide the fact ta



ity is bad

ood idea
eport

asures



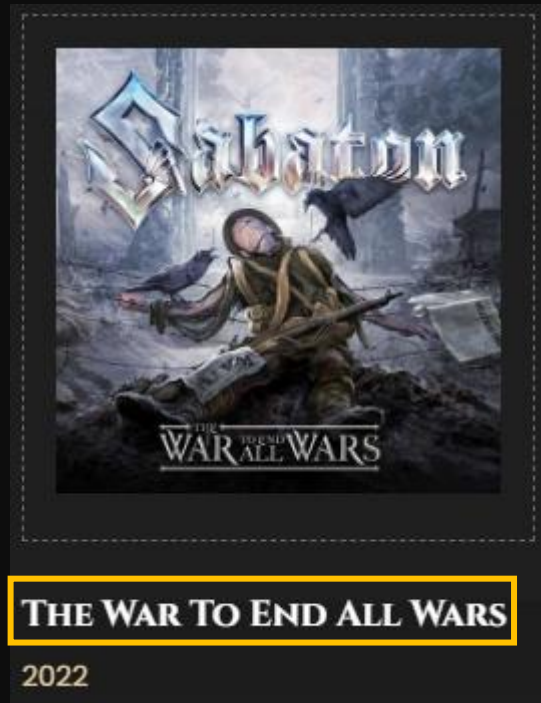
Column1

Sum of Column2

FOR ALL OF US WAITING

The Art Of War	2008
The Attack Of The Dead Men ft. Radio Tapok	2020
The Duelist	2025
Total	64582

Get data – Sabaton.net



sabaton.net/discography/the-war-to-end-all-wars/

THE WAR TO END ALL WARS 2022

▶ PLAY

DISC 1

1. Sarajevo
2. Stormtroopers
3. Dreadnought
4. The Unkillable Soldier
5. Soldier Of Heaven



Get data – Sabaton.net

The screenshot displays the Power BI Desktop environment. In the background, a data table is visible with the following content:

	Albums	Year
21	Coat Of	2006
22	The Art	2005
23	Metalize	2005
24	Attero Dominatus	2006
25	Primo Victoria	2005
26		

Overlaid on the table is a "Navigator" window showing a search bar and tabs for "Table View" and "Web View". Below the table, a dialog box titled "Add Table Using Examples" is open, featuring "OK" and "Cancel" buttons. To the right, the "Query Settings" pane is visible, showing the "PROPERTIES" section with the name "Albums" and the "APPLIED STEPS" section listing "Source", "Extracted Table From Html", and "Changed Type".

Get data – workings of a function

SongsOfAlbum

Display Options ▾

```
let
Source = Web.BrowserContents("https://www.sabaton.net/discography/" & Prm_Album & "/"),
#"Extracted Table From Html" = Html.Table(Source, {{"Column1", ".order-no"}, {"Column2", ".track-title"}, {"Column3", ".track-duration"}, {"Column4", ".btn-small"}}, [RowSelector=".track"]),
#"Changed Type" = Table.TransformColumnTypes("#Extracted Table From Html",{{"Column1", Int64.Type}, {"Column2", type text}, {"Column3", type time}, {"Column4", type text}})
in
#"Changed Type"
```

GetAlbumSongs [3]

-  SongsOfAlbum
-  Prm_Album (Carolus-Rex)
-  Fnc_GetAlbumSongs

Current Value

Carolus-Rex

Manage Parameter

Fnc_GetAlbumSongs

Display Options ▾

```
let
Source = (Album_Name as any) => let
Source = Web.BrowserContents("https://www.sabaton.net/discography/" & Album_Name & "/"),
#"Extracted Table From Html" = Html.Table(Source, {{"Column1", ".order-no"}, {"Column2", ".track-title"}, {"Column3", ".track-duration"}, {"Column4", ".btn-small"}}, [RowSelector=".track"]),
#"Changed Type" = Table.TransformColumnTypes("#Extracted Table From Html",{{"Column1", Int64.Type}, {"Column2", type text}, {"Column3", type time}, {"Column4", type text}})
in
#"Changed Type"
in
Source
```


Get data – Sabaton.net



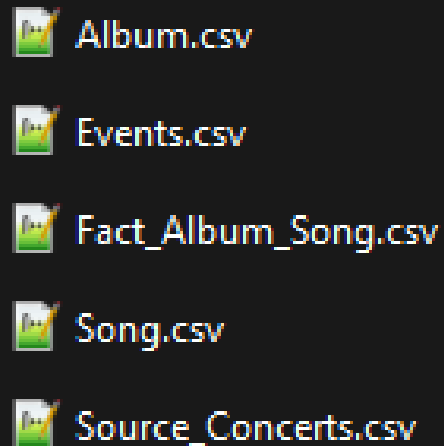
Column1	Column2	Func_GetAlbumSongs
Valid 100%	Valid 76%	Valid - %
Error 0%	Error 0%	Error 33%
Empty 0%	Empty 24%	Empty - %
Legends	2025	Table
Crossing the Rubicon (feat. Nothing More)	202	Error
The Duelist	2025	Table
Hordes of Khan	2025	Table
Templars	2025	Table
Stories From The Western Front	2023	Table
Heroes of The Great War	2023	Table
Weapons Of The Modern Age	2022	Table
The War To End All Wars	2022	Table
Steel Commanders	2021	Table
Kingdom Come	2021	Table
Defence Of Moscow	2021	Error
The Royal Guard	2021	Error
Livgardet	2021	Table
The Attack Of The Dead Men ft. Radio Tapok	2020	Table
Angels Calling ft. Apocalyptica	2020	Table
The Great War	2019	Table
Bismarck	2019	Table
The Last Stand	2016	Table
Heroes	2014	Table
Carolus Rex	2012	Table
Coat Of Arms	2010	Table
The Art Of War	2008	Table
Metalizer	2007	Table
Attero Dominatus	2006	Table
Primo Victoria	2005	Table

Get data – Sabaton.net

- Solve errors:
 - Not owning the source data:

```
#"Replaced Value1" = Table.ReplaceValue("#Replaced Value","Defence-Of-Moscow","Defence-Of-Moscow-2",Replacer.ReplaceText,{"Column1 - Copy"}),  
#"Replaced Value2" = Table.ReplaceValue("#Replaced Value1","The-Royal-Guard","The-Royal-Guard-2",Replacer.ReplaceText,{"Column1 - Copy"}),
```

- Save data
 - Owning the source data



A list of five CSV files, each preceded by a small green folder icon with a yellow arrow. The files are: Album.csv, Events.csv, Fact_Album_Song.csv, Song.csv, and Source_Concerts.csv.

- Album.csv
- Events.csv
- Fact_Album_Song.csv
- Song.csv
- Source_Concerts.csv

Get data

STAR SCHEMA



ALL THE THINGS

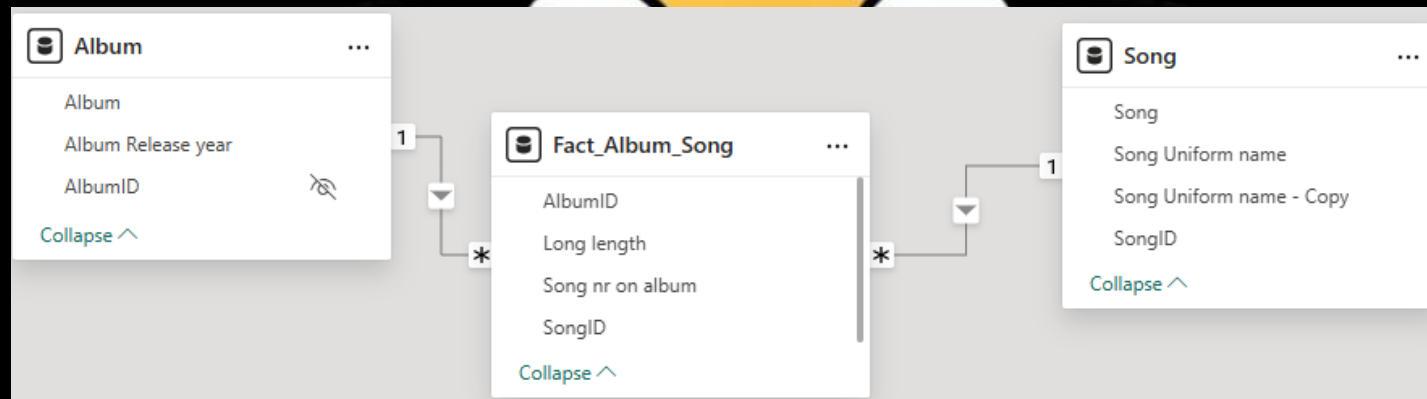
Get data

Dimensions

Fact table values

Album	Album Release year	Song nr on album	Song	Long length
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
Legends	2025	1	Templars	4:54:00
Legends	2025	2	Hordes of Khan	3:43:00
Legends	2025	3	A Tiger Among Dragons	4:06:00
Legends	2025	4	Crossing the Rubicon	3:29:00
Legends	2025	5	I, Emperor	4:16:00
Legends	2025	6	Maid of Steel	3:17:00
Legends	2025	7	Impaler	4:44:00
Legends	2025	8	Lightning at the Gates	4:12:00
Legends	2025	9	The Duelist	3:55:00
Legends	2025	10	The Cycle of Songs	5:39:00
Legends	2025	11	Till Seger	3:27:00
Crossing the Rubicon (feat. Nothing More)	2025	1	Crossing the Rubicon (feat. Nothing More)	3:24:00
The Duelist	2025	1	The Duelist	3:55:00
Hordes of Khan	2025	1	Hordes of Khan	3:43:00
Templars	2025	1	Templars	4:54:00
Stories From The Western Front	2023	1	1916	3:36:00
Stories From The Western Front	2023	2	Great War	4:28:00
Stories From The Western Front	2023	3	Hellfighters	3:26:00
Stories From The Western Front	2023	4	The Price Of a Mile	5:55:00

Get data



Text is not a good Key

			= Table.TransformColumns(#"Trimmed Text",{{"Song Uniform name - Copy", Text.Clean, type text}})	
--	--	--	---	--

ABC 123 Song	1 ² 3 SongID	A ^B C Song Uniform name	A ^B C Song Uniform name - Copy
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1 Templars	1 Templars	templars	
2 Hordes of Khan	2 Hordes of Khan	hordes of khan	
3 A Tiger Among Dragons	3 A Tiger Among Dragons	a tiger among dragons	
4 Crossing the Rubicon	4 Crossing the Rubicon	crossing the rubicon	
5 I, Emperor	5 I, Emperor	i, emperor	
6 Maid of Steel	6 Maid of Steel	maid of steel	
7 Impaler	7 Impaler	impaler	
8 Lightning at the Gates	8 Lightning at the Gates	lightning at the gates	
9 The Duelist	9 The Duelist	the duelist	
10 The Cycle of Songs	10 The Cycle of Songs	the cycle of songs	
11 Till Seger	11 Till Seger	till seger	
12 Crossing the Rubicon (feat. Nothing More)	12 Crossing the Rubicon	crossing the rubicon	
13 1916	13 1916	1916	
14 Great War	14 Great War	great war	
15 Hellfighters	15 Hellfighters	hellfighters	
16 The Price Of a Mile	16 The Price Of a Mile	the price of a mile	
17 Fields Of Verdun	17 Fields Of Verdun	fields of verdun	

Query Settings

PROPERTIES

Name

 Song

APPLIED STEPS

- Source
- Removed Other Columns
- Removed Duplicates
- Added Index
- Duplicated Column
- Extracted Text Before Delimiter
- Extracted Text Before Delimit...
- Extracted Text Before Delimit...
- Renamed Columns2
- Duplicated Column1
- Lowercased Text
- Trimmed Text
- X Cleaned Text**

Rework data

A ^B C Album	1 ² 3 Album Release year	ABC 123 Song nr on album	ABC 123 Song	ABC 123 Long length
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
Legends	2025	1	Templars	4:54:00
Legends	2025	2	Hordes of Khan	3:43:00
Legends	2025	3	A Tiger Among Dragons	4:06:00
Legends	2025	4	Crossing the Rubicon	3:29:00
Legends	2025	5	I, Emperor	4:16:00
Legends	2025	6	Maid of Steel	3:17:00
Legends	2025	7	Impaler	4:44:00
Legends	2025	8	Lightning at the Gates	4:12:00
Legends	2025	9	The Duelist	3:55:00

ABC 123 Song nr on album	ABC 123 Long length	1 ² 3 AlbumID	1 ² 3 SongID
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	4:54:00	1	1
1	4:54:00	5	1
2	3:43:00	1	2
1	3:43:00	4	2
3	4:06:00	1	3
4	3:29:00	1	4
5	4:16:00	1	5
6	3:17:00	1	6
7	4:44:00	1	7

Enrich the data – historical calendar

[News](#) [Tour](#) [VIP](#) [Music **New**](#) [Animated Movie](#) [Shop](#) [Video ▾](#) [Special ▾](#) [Historic Calendar](#) [Band ▾](#)

[f](#) [@](#) [t](#) [d](#) [v](#) [y](#) [e](#) [a](#) [s](#) [g](#) [n](#) [e](#)

HISTORICAL CALENDAR

THE STORY BEHIND EVERY SABATON SONG

[JAN](#) [FEB](#) [MAR](#) [APR](#) [MAY](#) [JUN](#) [JUL](#) [AUG](#) [SEP](#) [OCT](#) [NOV](#) [DEC](#)

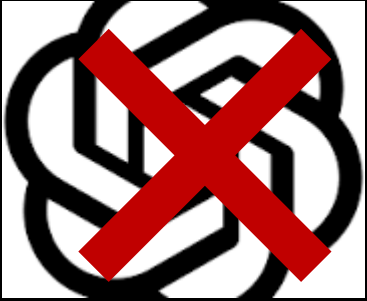


2 SEPTEMBER - 1964
82ND ALL THE WAY
Alvin York dies
Alvin York led the 82nd Infantry

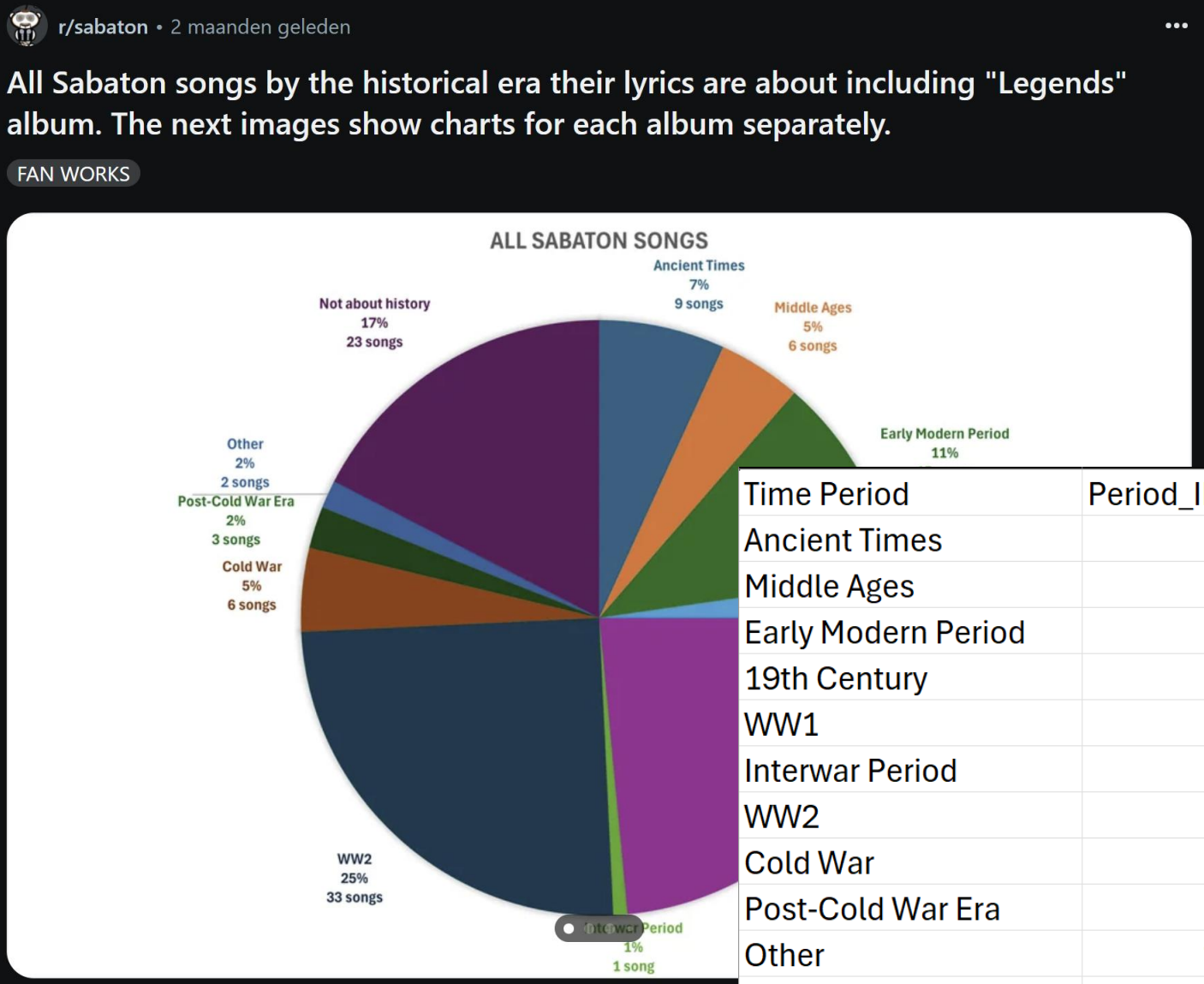


6 SEPTEMBER - 1974
METAL MACHINE

Enrich the data – Era's



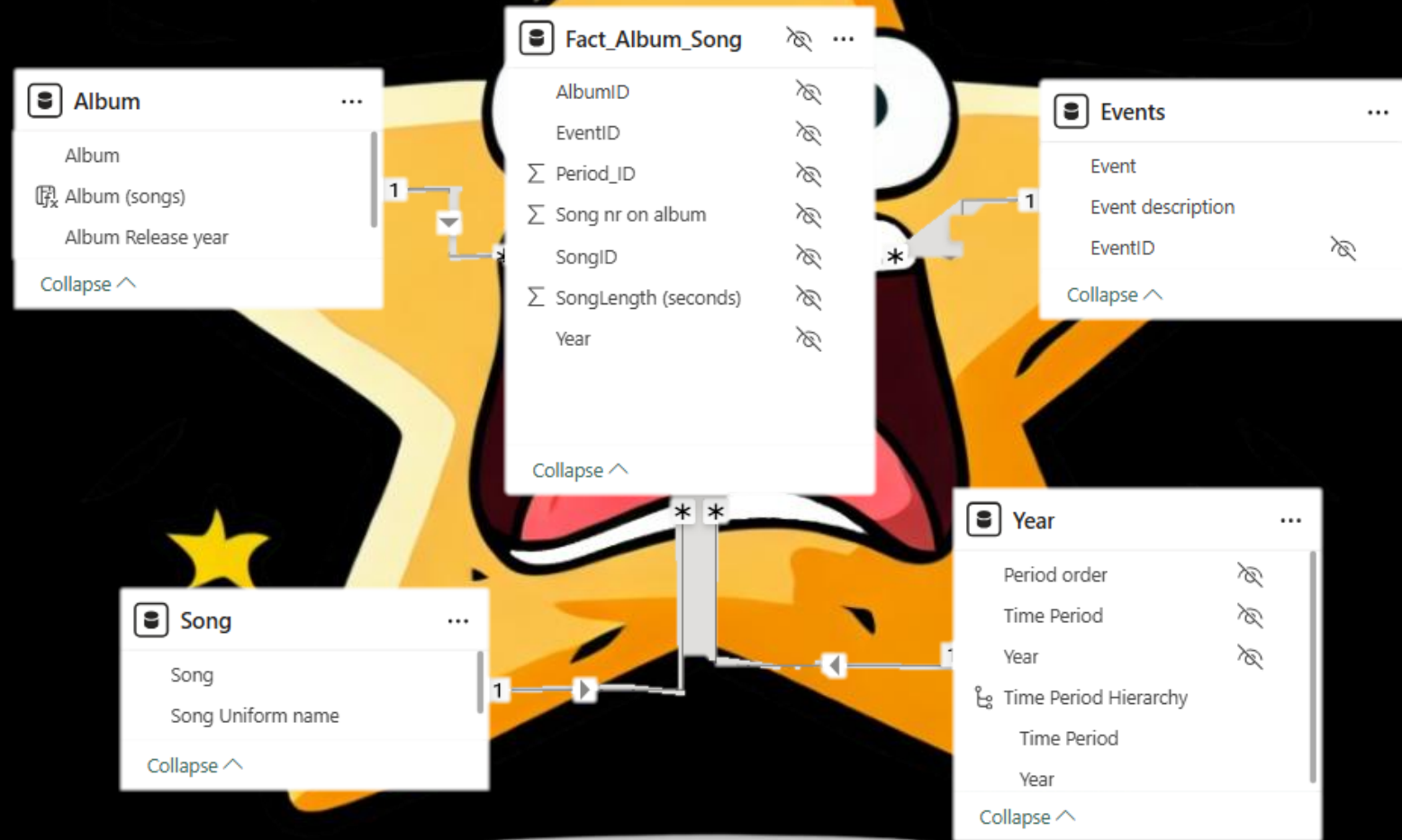
Enrich the data – Era's



Period_ID	Song
1	Sun Tzu Says
1	The Art of War
1	The Nature of Warfare
1	A Secret
1	Sparta
1	A Tiger Among Dragons
1	Crossing the Rubicon
1	Lightning at the Gates
1	The Cycle of Songs
2	Swedish Pagans
2	Blood of Bannockburn

Time Period	Period_ID	time frame start	time frame	Time order
Ancient Times	1	0	476	1
Middle Ages	2	477	1453	2
Early Modern Period	3	1454	1800	3
19th Century	4	1801	1913	4
WW1	5	1914	1918	5
Interwar Period	6	1919	1938	6
WW2	7	1939	1945	7
Cold War	8	1946	1989	8
Post-Cold War Era	9	1990	2025	9
Other	10			
Not about history	11			

Get data



Measures

>  'Measures table

DAX – language

- A Formula expression language
- Used to create measures
 - Saved in the model
 - Calculated on the fly
- Used to create calculated columns
 - Calculated values saved in the model

Measures

```
# albums = COUNTROWS(VALUES(Fact_Album_Song[AlbumID]))
```

```
# songs (wrong amount) = COUNTROWS(VALUES(Fact_Album_Song[SongID]))
```

```
average song durement =
```

```
var _Minutes = INT(AVERAGE(Fact_Album_Song[SongLength (seconds)])/60)  
var _Seconds = INT(MOD(AVERAGE(Fact_Album_Song[SongLength (seconds)]), 60))  
RETURN  
IF(SUM(Fact_Album_Song[SongLength (seconds)])>0,  
_Minutes & ":" & IF(_Seconds<10, 0) & _Seconds)
```

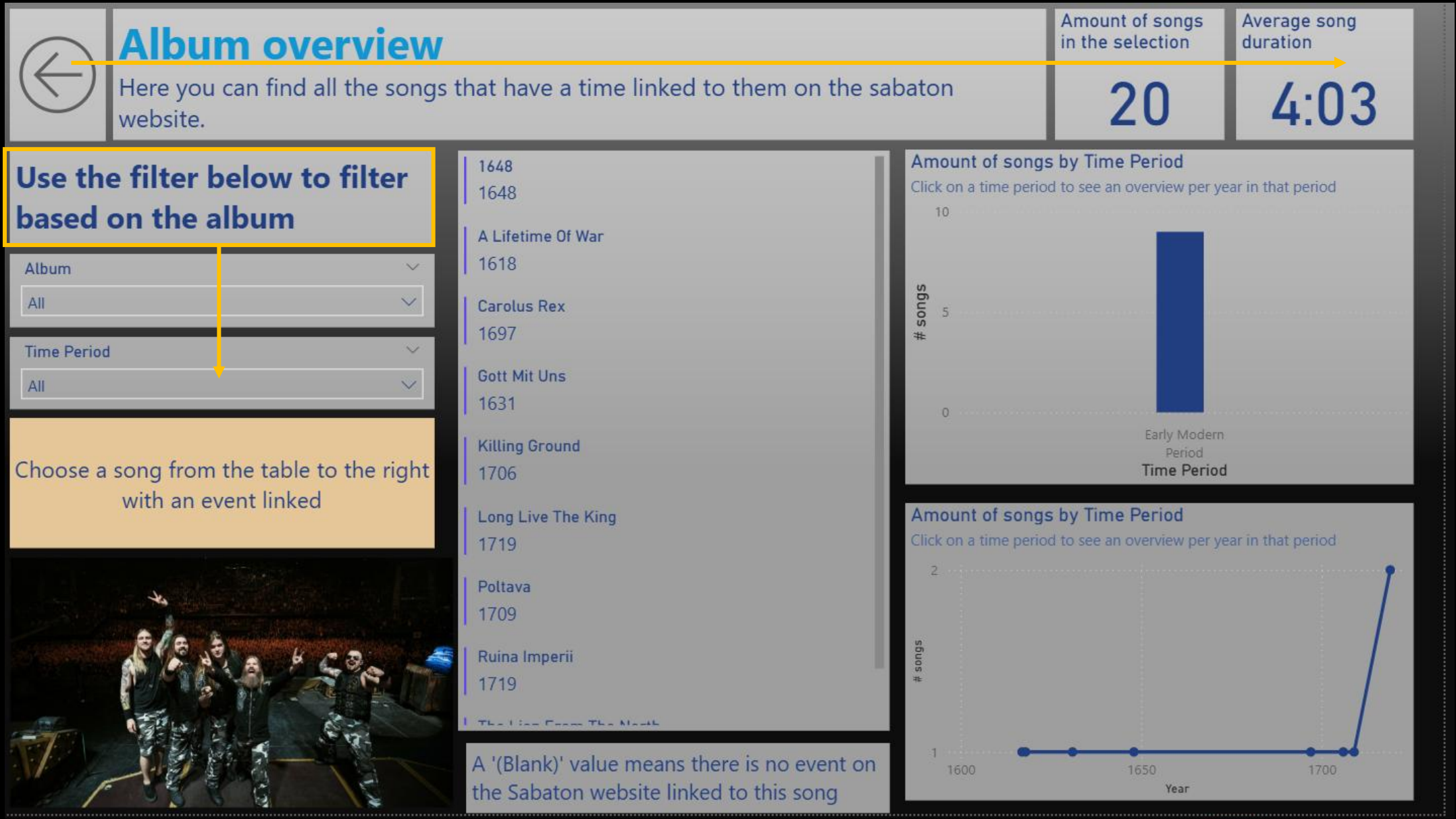
```
txt_btn_PublishYear =
```

```
"Click here for more details about albums released in " &  
SELECTEDVALUE(Album[Album Release year])
```

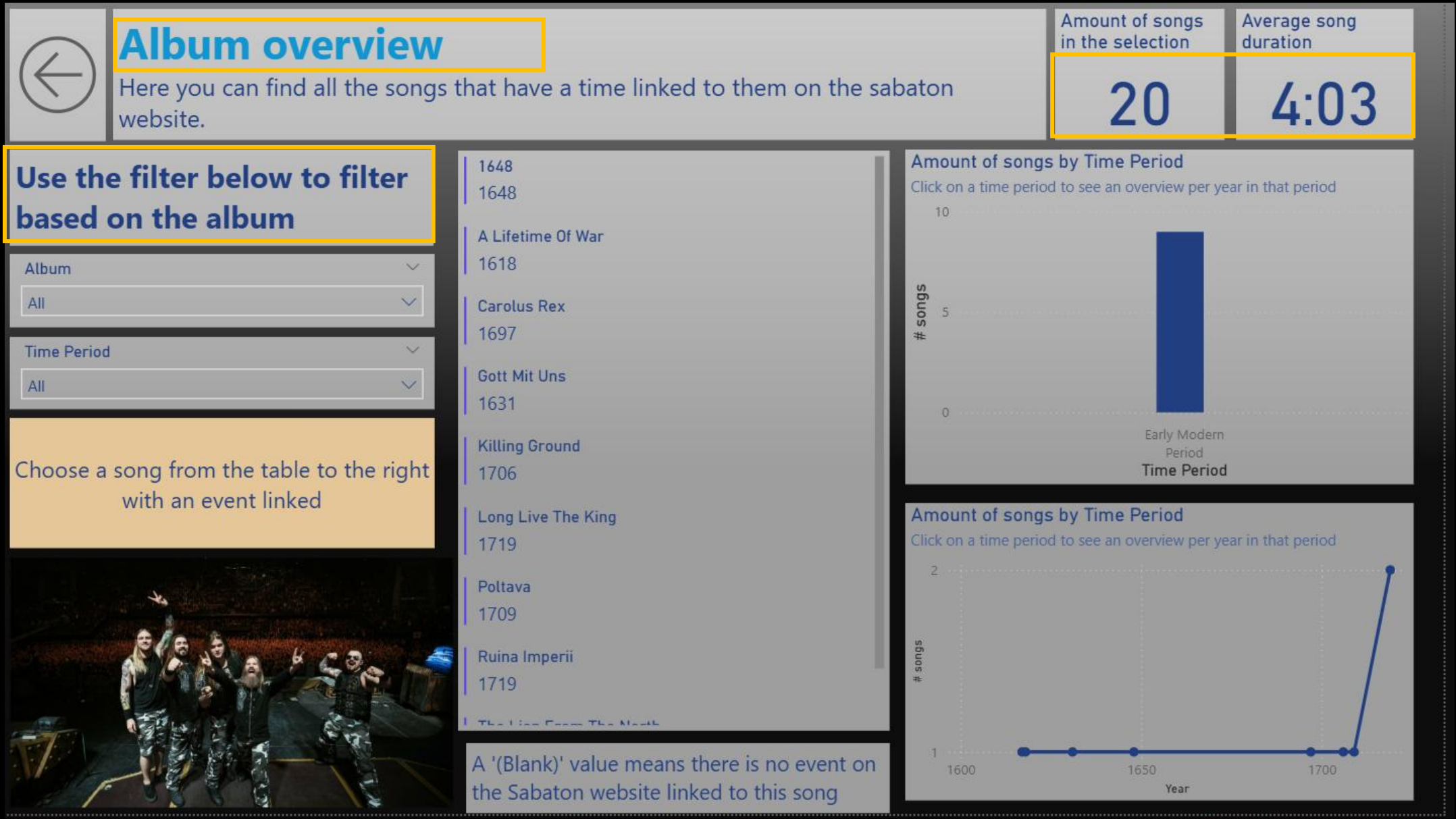
Visuals

- People start reading from the top left
- Grab attention
 - Big font
 - Colours
- Real estate is valuable
 - Including white spaces
- explain things via
 - Titles
 - Subtitles
 - Variable text
- Gestalt principles

Visuals - People start reading from the top left



Visuals - Grab attention



Amount of songs in the selection

20

Average song duration

4:03

Amount of songs by Time Period

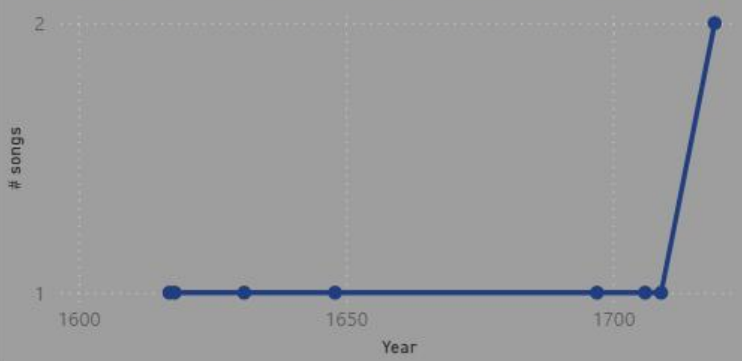
Click on a time period to see an overview per year in that period



Time Period	# songs
Early Modern Period	9

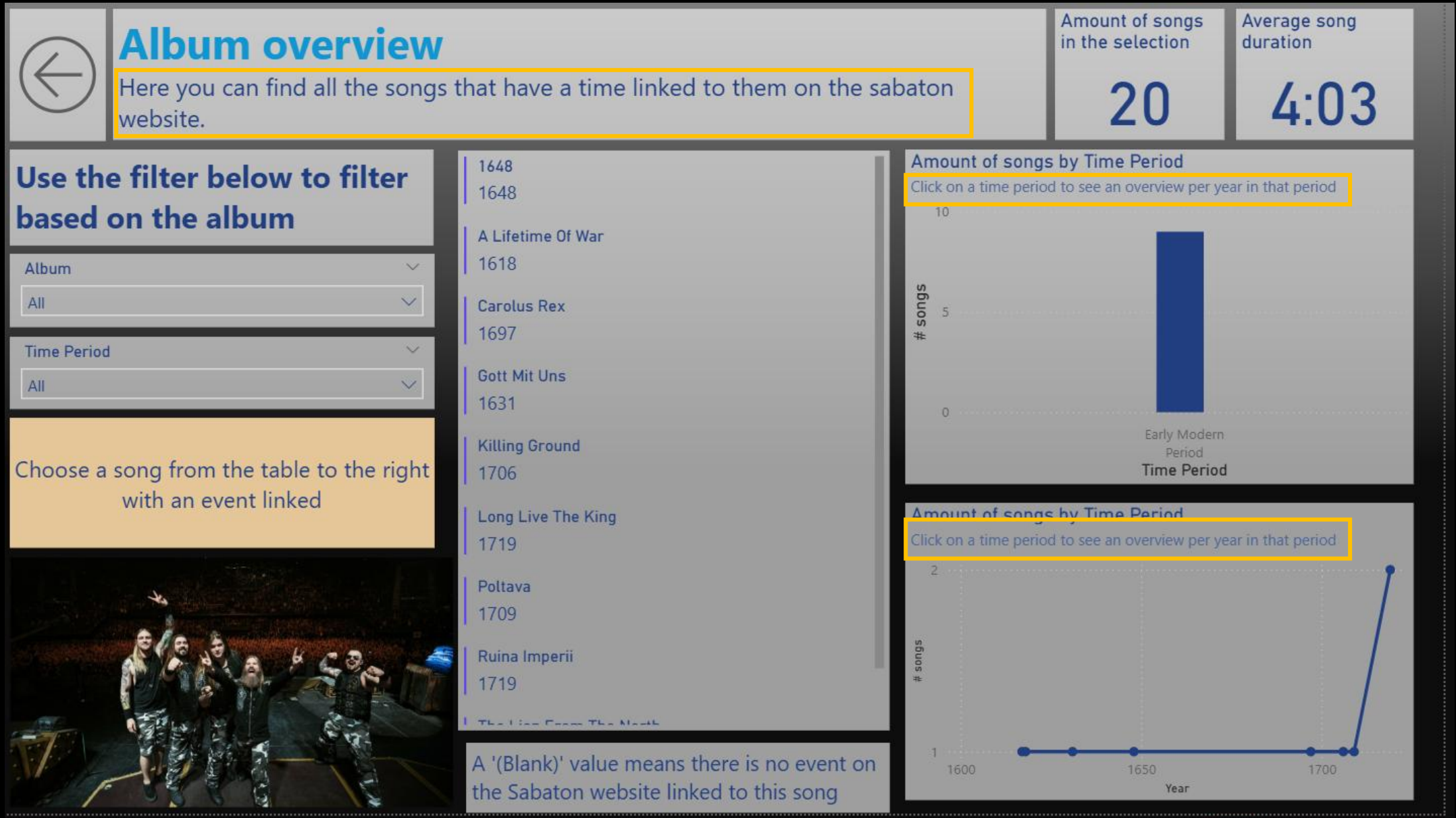
Amount of songs by Time Period

Click on a time period to see an overview per year in that period



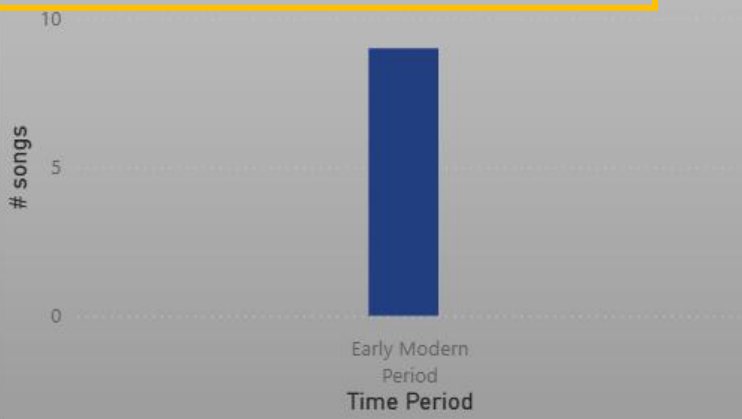
Year	# songs
1600	1
1618	1
1631	1
1648	1
1697	1
1706	1
1709	1
1719	2

Visuals - explain things



Amount of songs by Time Period

Click on a time period to see an overview per year in that period



songs

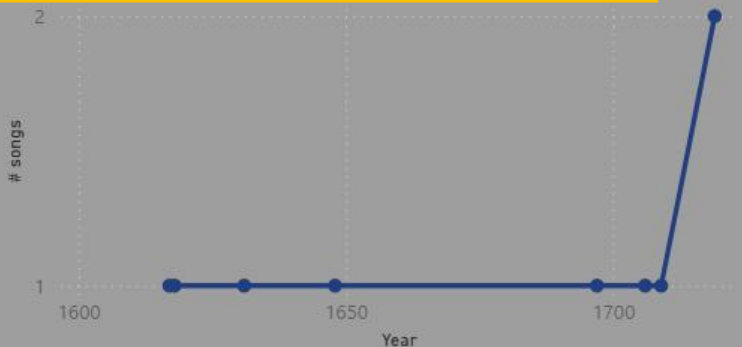
Early Modern Period

Time Period

Time Period	# songs
Early Modern Period	9

Amount of songs by Time Period

Click on a time period to see an overview per year in that period



songs

Year

Year	# songs
1600	1
1610	1
1620	1
1630	1
1640	1
1650	1
1660	1
1670	1
1680	1
1690	1
1700	1
1710	1
1720	2

Visuals – Real estate is valuable

